

Digital Image Watermarking with 1/T Rate Forward Error Correction to Malicious Attacks*

Wei-Hung Lin¹, Shi-Jinn Horng^{1,2} and Tzong-Wann Kao³, Yuh-Rau Wang⁴, Cheng-Ling Lee⁵
and Yi Pan⁶, *senior member of IEEE*

¹Dept. of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei, Taiwan, 106

²Dept. of Electronic Engineering, National United University, Miao-Li, Taiwan, 36003

³Dept. of Electronic Engineering, Technology and Science Institute of Northern Taiwan, Taipei, Taiwan

⁴Dept. of Computer Science and Information Engineering, St. John's University, Taipei, 25135 Taiwan

⁵Department of Electro-Optical Engineering, National United University, Miaoli, Taiwan

⁶Dept. of Computer Science, Georgia State University, Atlanta, GA 30302

Email: ¹ weber3013@yahoo.com.tw, ² horngsj@yahoo.com.tw, ³ tkao@tsint.edu.tw,
⁴ yrwang@mail.sju.edu.tw, ⁵ cherry@nuu.edu.tw, ⁶ pan@cs.gsu.edu

Abstract

In this paper, a copyright protection method for digital image with 1/T rate forward error correction (FEC) is proposed. In the proposed method, the original image is lossless and the watermark is robust to malicious attacks including geometric attacks such as scaling, rotation, cropping, print-photocopy-scan, and scaling-cropping attacks and nongeometric attacks such as low-pass filtering, sharpening, JPEG compression attacks. The watermark logo is fused with noise bits to improve the security, and later XORed with the feature value of the image by 1/T rate FEC. During extraction, the watermark bits are determined by majority voting, and the extraction procedure needs neither the original image nor the watermark logo. Experimental results show that not only the image is lossless but also the proposed method can effectively resist the common geometric and nongeometric attacks. Since the proposed method is based on spatial domain and transferring to frequency domain is not needed, the embedding and extraction performances are improved.

1 Introduction

The Internet has brought convenience and innovation. Most users use it to exchange information. Owing to publicity about intellectual

property rights, many researchers are aware of the issues of copyright protection, image authentication, proof of ownership, etc; hence, there are many solutions that have been proposed. The watermarking technique is one of the solutions. This technique embeds information so that it is not easily perceptible; that is, the viewer cannot see any information embedded in the contents. There are several important essential properties in the watermarking system. First is transparency. The embedded watermark should not degrade the quality of the image and should be perceptually invisible to maintain its protective secrecy. Second is robustness. The watermark must be robust enough to resist common image processing attacks (blurring, sharpening, noising, JPEG compression, etc.) and geometric attacks (scaling, cropping, rotation, print-photocopy-scan, etc.), and must not be easily removable. Third is security. Only the owner of the image is able to extract the watermark. Fourth is clarity. The identification of the owner of the image cannot be ambiguous. Fifth is blind reproducibility. It may only need secrecy keys [1], or even need neither the original image nor original watermark for extraction. The last is multiple watermarking. An embedded watermark cannot interfere with any other watermark which has been embedded in the image.

Many watermarking methods have been proposed. Cox et al. [2, 3] embedded the watermark into the perceptually significant portion of the whole DCT-transformed image wherein a predetermined range of low frequency components excludes the DC component. This watermarking scheme has been shown to be robust against common attacks, such as compression, filtering, cropping, etc. Podilchuk and Zeng [4]

* This work was supported in part by the National Science Council under contract number NSC 96-2918-I-011-002, 96-2221-E-011-022-, 95-2221-E-011-032-MY3.
Contact author: Shi-Jinn Horng, Email: horngsj@yahoo.com.tw

improved Cox's method and added the model of the just noticeable difference (JND) to select the maximum length and maximum power watermark sequence. In [5-12], the watermark was embedded by quantizing the coefficients in the DCT-block. Langelaar et al. [13] proposed a blind watermarking approach called differential energy watermarking. A set of several 8×8 DCT blocks are composed and divided into two parts to embed a watermark bit. The high frequency DCT coefficients in the JPEG/MPEG stream are selectively discarded to produce energy differences in the two parts of the same set.

Kimpan et al. [14] proposed a watermarking method for still images in the spatial domain. The watermark logo is embedded in the original image by modifying the gray levels of pixels in the original image block to appropriate an intensity of a block. A semi-blind watermark was proposed by Mahmood and Selin [15]. The watermark is embedded by a spatially adaptive wavelet threshold method to select the coefficients.

Huang and Yang [16] proposed a watermarking algorithm based on the DWT. The original image is separated into m blocks, each of size $n \times n$; then every block is decomposed into a wavelet domain. The watermark is embedded in the wavelet coefficients in the middle and low subbands of a block in each image. Khelifi et al. [17] proposed an adaptive blind watermarking technique based on the DWT. The original image is separated into non-overlapping blocks classified as uniform or non uniform blocks using a JND-based classifier. The watermark is embedded in the high subband of each block which is transformed into the DWT according to its classification. Zhang et al. [18] divided the original image into $n \times n$ blocks and transformed them into a DWT domain. The watermark is embedded by using the mean and the variance of a subband to modify the wavelet coefficient of a block.

Lien et al. [19] proposed a wavelet-based blind watermarking scheme. The wavelet coefficients of the host image are grouped into wavelet trees, and each two watermark bits are embedded using four trees. According to the binary value of two watermark bits which are embedded, one of the four trees is quantized with respect to a quantization index, and four trees exhibit a large statistical difference between the quantized tree and the unquantized tree; the difference can later be used for watermark extraction. For the existing methods listed in [2-19], the original image is lossy since the original coefficient is modified by quantizing the coefficient of the image in the embedding process. Moreover, these methods are not strong enough to resist geometric attacks, such as rotation and print-photocopy-scan. Chen et al.

[20] proposed a blind and lossless wavelet-based copyright protection scheme. The wavelet coefficients of the low frequency are compared with the average coefficient of the low frequency to generate the feature value. Then the feature value is XORed with watermark bits to generate the verification key to the trusted third party, which is called a certification authority. The verification process (through the trusted third party) arrives at non-repudiation of the extracted watermark. Lou et al. [21] improved Chen et al's method by adding the torus automorphism to scramble the original image to promote security.

Packet losses or packet errors often occur in data communication. Forward error correction (FEC) is an error correction technique to compensate packet loss in the Internet [22-31]; especially in real-time multimedia which has a strict delay requirement, such as video transmission [22, 23, 28], and voice or audio application [25]. FEC is also applied in wireless communication [26, 28, 29, 32], since wireless communications, which usually are with higher packet loss rates and lower transmission rates, are more unreliable than transmissions by wire communications. FEC adds the redundant information to the packet in a transmission to achieve the objective of error correction and high reliability. In the Bluetooth technique, the header information is protected with a $1/3$ rate forward error correction [33-35], and the receiver subsequently uses majority voting to decide the value of the bit which has been transmitted. The $1/3$ rate forward error correction means that one bit is transmitted thrice.

Digital signature can ascertain the source and content of information via auditing and tracing back. It can provide integrity, non-repudiation, authentication, and confidentiality. Timestamp can trace back and provide a time certification by trusted third party [36]. By the digital signature and timestamp, the digital content can be effectively protected in copyright protection.

In this paper, we propose a copyright protection scheme using the $1/T$ rate FEC for resisting malicious attacks, where T is the times of data redundancy. The feature value, which we call the B-map (binary map) table, is created and is the same as that in [20]; subsequently the B-map is divided into T blocks. The watermark logo is spread over all the blocks of the image, whose size is the total size of the watermark logo and the noise bits. We generate the protection key through the watermark logo fused with the noise bits and XORed with the feature value of the image by the $1/T$ rate FEC. In the verification process, the extraction of the watermark needs neither the original image nor the watermark logo, and the

watermark bits are determined by majority voting. The non-repudiation of the extracted watermark uses the protection key stored in the trusted third party to verify whether the image is legal or not. Since the proposed method is based on the spatial domain, it can reduce the computation time when the image in the spatial domain is transformed to the frequency domain. Experimental results show that the proposed method can effectively resist common geometric and nongeometric attacks.

This paper is organized as follows: The image protection and verification using the 1/T forward error correction are described in Section 2. The experimental results and discussion are given in Section 3. Finally, the conclusions are summed up in Section 4.

2 THE PROPOSED METHOD

In the propose method, we do not transfer the protected images to frequency domain, and hence it can reduce the computation time of embedding and extraction process. In order to raise secrecy and confusion of a watermark, and to make the embedded information more difficult to be detected by attackers, noise bits are used. After embedding, the proposed watermark scheme will generate a verification key which will be used to register to a trusted third party with digital signature and time stamp. The register process can provide integrity, non-repudiation, authentication, and confidentiality. If there is any dispute, the digital signature and time stamp can be used to verify the copyright image and watermark. Since the error correction via 1/T rate and the watermark bits are distributed everywhere in an image, the watermark can be extracted accurately. The proposed watermarking scheme is lossless and robust for application in copyright protection. The detailed description of algorithms for the image protection and verification are is stated as follows:

A. Image protection generation algorithm

Assume an original image I is a gray level image (8 bits/pixel), watermark logo W is a binary image, and noise energy E is a binary value. They are defined as follows:

$$I = \{i_{x,y} \mid 0 \leq i_{x,y} \leq 255, 0 \leq x < I_W, 0 \leq y < I_H\}, \quad (1)$$

where $i_{x,y}$ is the gray level coefficient located at coordinate (x, y) of the original image. I_W and I_H are the width and height of image I , respectively.

$$W = \{w_{m,n} \mid w_{m,n} \in \{0,1\}, 0 \leq m < W_W, 0 \leq n < W_H\}, \quad (2)$$

where $w_{m,n}$ is the coefficient located at coordinate (m, n) of the watermark logo. W_W and W_H are the width and height of watermark logo W ,

respectively.

$$E = \{e_{m,n} \mid e_{m,n} \in \{0,1\}, 0 \leq m < W_W, 0 \leq n < W_H\}, \quad (3)$$

where each $e_{m,n}$ is chosen independently according to $N(0,1)$ (here $N(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2).

Step 1. Permute the watermark logo W by Eqs. (4) and (5),

$$W^{new} = \left\{ \begin{array}{l} w_{m,n}^{new} \mid w_{m,n}^{new} = \text{Perm}_{s1}(w_{m,n}), \\ 0 \leq m \leq W_W, 0 \leq n \leq W_H \end{array} \right\}, \quad (4)$$

where $\text{Perm}_{s1}(\cdot)$ denotes a pseudorandom permutation function with seed $s1$.

Step 2. Fuse W^{new} with noise E to generate WE by

$$WE = W^{new} \parallel E, \quad (5)$$

where

$$WE = \left\{ \begin{array}{l} we_{m,n}^{new} \mid we_{m,n}^{new} = w_{m,n}^{new}, 0 \leq m < W_W, \\ 0 \leq n < W_H, \text{ and } we_{m,n}^{new} = e_{m,n}, \\ W_W \leq m < 2W_W, 0 \leq n < W_H, \end{array} \right\}$$

$$WE_{new} = \text{Perm}_{s2}(WE), \quad (6)$$

where $\parallel$ denotes an union operation.

Perm_{s2} denotes the pseudorandom permutation function with seed $s2$. The size of WE is equal to the size of W^{new} added to the size of E .

In this step, in order to increase the secrecy of the watermark, the noise E is added to the watermark and then we shuffle them using seed $s2$.

Step 3. Calculate the average gray level value $\text{avg}(I)$ of the image I and generate the feature value to B-map table B ,

$$\text{avg}(I) = \frac{1}{I_W \times I_H} \sum_{x=0}^{I_W-1} \sum_{y=0}^{I_H-1} i_{x,y} \quad (7)$$

$$B = \left\{ \begin{array}{l} b_{x,y} \mid b_{x,y} \in \{0,1\}, \\ 0 \leq x < I_W, 0 \leq y < I_H \end{array} \right\}, \quad (8)$$

$$\text{where } b_{x,y} = \begin{cases} 1 & \text{if } i_{x,y} > \text{avg}(I) \\ 0 & \text{if } i_{x,y} \leq \text{avg}(I) \end{cases}.$$

Step 4. Divide B into G blocks.

In this step, we calculate the maximum number of blocks of size WE^{new} can be embedded in an image. According to the size of WE^{new} , B is separated into G blocks by WE^{new} .

$$G = \left\lfloor \frac{I_W}{2 \times W_W} \right\rfloor \times \left\lfloor \frac{I_H}{W_H} \right\rfloor, \quad (9)$$

$$Block_t = \left\{ \begin{array}{l} b_{x,y} \mid b_{x,y} \in \{0,1\}, \\ 2 \times W_w \times (t-1) \leq x \leq 2 \times W_w \times t-1, \\ W_H \times (t-1) \leq y \leq W_H \times t-1, \\ 1 \leq t \leq T, 1 \leq T \leq G \end{array} \right\}, \quad (10)$$

where the size of a block is the same as that of WE^{new} . Since the width of WE^{new} equals to the width of W_w add the width of E and the width of W_w equals to the width of E , here we use $2 \times W_w$ to denote the width of WE^{new} .

Step 5. Generate the verification key by the $1/T$ rate FEC

$$K = FEC(T, Block_t \oplus WE^{new}), 1 \leq t \leq T, \quad (11)$$

where $\oplus$ denotes the XOR operation, $FEC(.)$ denotes WE^{new} is embedded in the t -th $Block$ (i.e., $Block_t$) by the XOR operation for $1 \leq t \leq T$. Note that the size of K is still $I_W \times I_H$.

Step 6. Add owner digital signature to verification key and register S to a trusted third party. Using the larger T , the error can be decreased in the extraction process when the watermarked image is seriously destroyed by the attacks such as cropping, rotation, and Gaussian noise. In order to decrease the size of S , a compress function is used. Since the S consists of zero and one in ASCII format, the S can be lossless and can be effectively compressed. For example, in this case, if an image size is $512 \times 512 \times 8$ pixels (256Kbytes), the largest size of S is the same as the watermarked image. By using the Zip compression, the S can be decreased to about 31Kbytes.

$$Ck = C(Perm_{s3}(K)), \quad (12)$$

$$S = Sig_p(Ck, s1, s2, s3), \quad (13)$$

where $C(.)$ denotes a compression function, $Perm_{s3}$ denotes the pseudorandom permutation function with seed $s3$. $Sig_p(.)$ denotes a digital signature function using owner private key K .

Step 7. Timestamp to S .

Owner registers the S to a trusted third party. Subsequently, the third party generates a timestamp and appends to the S . If there is any dispute, the S and TSp can be

used to verify the validity of key K .

$$T = TSp(S). \quad (14)$$

Here TSp denotes timestamp function using third party's private key.

C. Image Verification algorithm

In the verification process, no original image is needed. We use third party's public key to verify the timestamp TS and use the owner's public key to verify the signature S , and get the $s1$, $s2$, $s3$ and compressed Ck . Then the verification step is as follows:

Step 1. Extract the verification key K .

$$K = DCom(Perm_{s3}(Ck)^{-1}), \quad (15)$$

where

$$K = \{k_{x,y} \mid k_{x,y} \in \{0,1\}, 0 \leq x < I_W, 0 \leq y < I_H\};$$

$DCom(.)$ denotes a decompress function, which is the inverse function of $C(.)$;

$Perm_{s3}^{-1}(.)$ denotes the inverse function of $Perm_{s3}(.)$ with seed $s3$.

Step 2. Get the attacked image I' and calculate the average gray level value $avg(I')$ of the attacked image I' ,

$$I' = \left\{ \begin{array}{l} i'_{x,y} \mid 0 \leq i'_{x,y} \leq 255, \\ 0 \leq x < I_W, 0 \leq y < I_H \end{array} \right\}, \quad (16)$$

$$avg(I') = \frac{1}{I_W \times I_H} \sum_{x=0}^{I_W-1} \sum_{y=0}^{I_H-1} i'_{x,y}. \quad (17)$$

Step 3. Generate the feature value B-map table B' ,

$$B' = \left\{ \begin{array}{l} b'_{x,y} \mid b'_{x,y} \in \{0,1\}, \\ 0 \leq x < I_W, 0 \leq y < I_H \end{array} \right\}, \quad (18)$$

$$\text{where } b'_{x,y} = \begin{cases} 1 & \text{if } i'_{x,y} > avg(I') \\ 0 & \text{if } i'_{x,y} \leq avg(I') \end{cases}.$$

Step 4. Extract the verification key WEG' by

$$WEG' = B' \oplus K, \quad (19)$$

$$\text{where } WEG' = \left\{ \begin{array}{l} weg'_{x,y} \mid weg'_{x,y} \in \{0,1\}, \\ 0 \leq x < I_W, 0 \leq y < I_H \end{array} \right\}.$$

Step 5. Divide the WEG' into G blocks

$$Block'_t = \left\{ \begin{array}{l} weg'_{x,y} \mid weg'_{x,y} \in \{0,1\}, \\ 2 \times W_w \times (t-1) \leq x \leq 2 \times W_w \times t-1, \\ W_H \times (t-1) \leq y \leq W_H \times t-1, \\ 1 \leq t \leq T, 1 \leq T \leq G \end{array} \right\}, \quad (20)$$

$$Block_i^p = Perm_{s2}(Block'_i)^{-1}, 1 \leq t \leq T, \quad (21)$$

where the value of T and the size of a block are the same as those defined in Eqs. (9) and (10). $\text{Perm}_{s2}(\cdot)^{-1}$ denotes the inverse function of $\text{Perm}_{s2}(\cdot)$ with seed $s2$.

Step 6. Use the majority vote to decide whether WE' is bit 1 or bit 0.

$$WE' = \text{MajorV}(Block_t^P), 1 \leq t \leq T. \quad (22)$$

$$\text{where } WE' = \left\{ \begin{array}{l} we'_{m,n} \mid we'_{m,n} \in \{0,1\}, \\ 0 < m < 2 \times W_W, 0 \leq n < W_H \end{array} \right\},$$

the MajorV function determines the maximum number of bit 0 and bit 1 of the same position of $Block_t^P$, $1 \leq t \leq T$. For example, the number of bit 1 at position (1,1) of every block is 10, and the number of bit 0 at position (1,1) of every block is 22. After MajorV function, $we'_{1,1}$ is turned out to be bit 0.

Step 7. Split the watermark bits and noise bits from WE' to get watermark bits.

$$W^e = \text{Split}(WE'). \quad (23)$$

The function Split(.) is the inverse function of Eq. (5).

Step 8. Inverse W^e to form the watermark logo W' .

$$W' = \text{Perm}_{s1}(W^e)^{-1}. \quad (24)$$

where $\text{Perm}_{s1}(\cdot)^{-1}$ denotes the inverse function of $\text{Perm}_{s1}(\cdot)$ with seed $s1$.

3 EXPERIMENTAL RESULTS AND DISCUSSIONS

A. Experiment results

Like other researchers, we use the peak signal-to-noise ratio (PSNR) to evaluate the quality between an attacked image and the original image. For the sake of completeness, the PSNR is formulated as follows:

$$PSNR = 10 \times \log_{10} \frac{255 \times 255}{\frac{1}{I_H \times I_W} \sum_{x=0}^{I_H-1} \sum_{y=0}^{I_W-1} [i(x,y) - i'(x,y)]^2} dB, \quad (25)$$

where I_H and I_W are the height and width of the image, respectively, and $i_{x,y}$ and $i'_{x,y}$ are the grey level coefficients located at coordinate (x, y) of the original image I and the attacked image I' , respectively.

After extracting the watermark, the normalized correlation coefficient (NC) is computed using the original watermark and the extracted watermark to judge the existence of the watermark in order to

measure the correctness of an extracted watermark. It is defined as follows:

$$NC = \frac{1}{W_H \times W_W} \sum_{i=0}^{W_H-1} \sum_{j=0}^{W_W-1} w_{i,j} \times w'_{i,j}, \quad (26)$$

where W_H and W_W are the height and width of the watermark. $w_{i,j}$ and $w'_{i,j}$ are the values located at coordinate (x, y) of the original watermark W and the extracted watermark W' . Here $w_{i,j}$ is set to 1 if it is watermark bit 1; otherwise, it is set to -1. $w'_{i,j}$ is set in the same way. So the value of $w_{i,j} \times w'_{i,j}$ is either -1 or 1.

We use the Lena image (512×512 pixels, 8 bits/pixel) as the test image in our experiments. The size of the binary watermark logo is 64×64, and the value of T is 32. For attacking, we use the StirMark benchmark [37] and PhotoImpact 11 software [38] tools to simulate common image attacks; the original Lena image and binary watermark are shown in Fig. 1.



Fig. 1. (a) The original image of Lena of size 512×512. (b) The original binary watermark of size 64×64.

In the following experiments, we consider both geometric and nongeometric attacks.

Nongeometric attacks include JPEG compression, low pass filtering, histogram equalization, and sharpening. The experimental results show that the watermark logo can still be recognized clearly.

Table 1. Normalized correlation coefficients (NC) upon attacks of JPEG compression, median filtering, and blurring.

	JPEG	Median filtering	Blurring
Attacked image			
PSNR	22	22	20
Retrieved logo	NTUST CSIE	NTUST CSIE	NTUST CSIE
NC	1	1	1

JPEG is one of the most used formats in the Internet and digital cameras. The JPEG quality factor is a number between 1 and 100 and

associates a numerical value with a particular compression level. When the quality factor is decreased from 100, the image compression is improved, but the quality of the resulting image is significantly reduced. In Table 1, the quality factors of the JPEG compression is 1, and we can also extract the watermark logo completely.

For other non-geometric attacks [37], there are median filtering, Blurring, sharpening, and histogram equalization. After these attacks, the resulting images are blurred or sharpened on the edge [39]; the proposed method is robust in resisting attacks and correctly extracts the watermark logo.

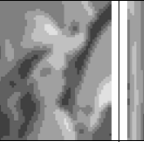
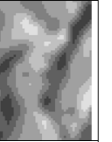
Table 2. Normalized correlation coefficients (NC) upon attacks of by sharpening, histogram equalization, and scaling.

	Sharpening	Histogram equalization	Scaling
Attacked image			
PSNR	14	19	17
Retrieved logo	NTUST CSIE	NTUST CSIE	NTUST CSIE
NC	1	1	1

Table 3. Normalized correlation coefficients (NC) upon attacks of rotation, noising, and cropping.

	Rotation	Noising	Cropping
Attacked image			
PSNR	10	9	11
Retrieved logo	NTUST CSIE	NTUST CSIE	NTUST CSIE
NC	0.85	0.81	0.98

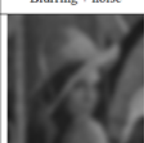
Table 4. Normalized correlation coefficients (NC) upon attacks of print-photocopy-scan, blurring + JPEG, and scaling + JPEG.

	Print-photocopy-scan	Blurring + JPEG	Scaling + JPEG
Attacked image			
PSNR	9	19	16
Retrieved logo	NTUST CSIE	NTUST CSIE	NTUST CSIE
NC	1	1	0.99

We also use other attack methods such as rotation, scaling, Gaussian noise, print-photocopy-scan, and cropping to conduct geometric attacks. For the rotation attack, the image is rotated at 6° [37]. For the scaling attack, an image of size 512×512 is

first scaled to 16×16 via PhotoImpact 11 software, then the scaled image is opened and resized to 512×512 . For the Gaussian noise attack, the variance of noise is 15; for print-photocopy-scan, we print the Lena image using a laser printer (HP LaserJet 9000); then the image which was photocopied is rescanned by a scanner within 75 dpi and 256 gray-level and resized to 512×512 pixels. For the cropping attack, an image of $1/4$ size is cropped via PhotoImpact 11 software.

Table 5. Normalized correlation coefficients (NC) upon attacks of blurring + noise, print-photocopy-scan + sharpening, histogram equalization, and scaling-cropping.

	Blurring + noise	Print-photocopy-scan + sharpening	Scaling-cropping
Attacked image			
PSNR	13	9	13
Retrieved logo	NTUST CSIE	NTUST CSIE	NTUST CSIE
NC	1	0.99	0.82

We combine various kinds of attacks in our experiments to test our proposed method, such as scaling + cropping (the test image is scaled from 512×512 to 670×670 pixels, then we cut the edge area of the test image to form the 512×512 test image); we also combine blurring + noise, blurring + JPEG, print-photocopy-scan + sharpening, print-photocopy-scan + scaling, and scaling + JPEG. After these attacks, the retrieved watermark logo can still be recognized clearly.

In Table 6, we compare the proposed method to Chen *et al.*'s [20] and Lou *et al.*'s [21] methods to demonstrate the robustness of our method. In the experiment, the Lena image (512×512 pixels, 8 bits/pixel) is used. The watermarked image is attacked (by blurring, JPEG, Noise, sharpening, print-photocopy-scan, and scaling attacks) more seriously than that in Chen *et al.*'s and Lou *et al.*'s methods; although our PSNRs are lesser than theirs, the value of NC under each attack is still 1. In other attacks (rotation, cropping, and cropping + scaling), all parameters are set to the same as those in Chen *et al.*'s method; that is, for the rotation attack, the image is rotated at 2° , for the cropping attack, the image is cropped by $1/4$, and for cropping + scaling attack, the image is scaled from 512×512 pixels to 560×560 pixels, then the edge area of the test image is cut to form the 512×512 test image. As you can see, the values of PSNRs of our method are quite closer to those in Chen *et al.*'s method; the value of NC under each attack is always higher than that of Chen *et al.*'s and Lou *et al.*'s methods. Note that all parameters of latter attacks such as rotation, cropping, and cropping + scaling are

not described in [21]. For cropping attack, the image was cropped by 3/4, the PSNR for the remaining image is still greater than 5; we are curious about the parameters set by Lou *et al.*'s method.

Table 6. Comparison with Lou *et al.*'s[21] and Chen *et al.*'s [20] methods.

Attack	Lou et al.'s method		Chen et al.'s method		Proposed method	
	PSNR	NC	PSNR	NC	PSNR	NC
Blurring	25	0.99	29	0.99	20	1
JPEG	24	0.98	31	0.98	22	1
Noise	28	0.99	30	0.99	16	1
Sharpening	28	0.99	28	0.99	14	1
Scaling	21	0.98	29	0.99	20	1
Rotation	8	0.93	14	0.82	13	0.99
Print-photocopy-scan	7	0.94	19	0.90	9	1
Cropping	5	0.92	11	0.87	11	0.98
Cropping+scaling	9	0.93	16	0.80	15	1

B. Discussions

The proposed method satisfies the properties of the watermarking including blindness, security, invisibility, robustness, unambiguity and transparency. Since the original image is lossless, multiple logos can be handled [20]. The proposed method can also be applied to the color image. The watermark bits can be embedded based on the RGB color values of the image. Furthermore, the proposed method is based on the spatial domain; it can reduce the computation time in which the image in the spatial domain transforms to the frequency domain. Hence, the proposed method can be applied to the video stream.

4 Conclusion

In this paper, copyright protection based on $1/T$ forward error correction for digital images is proposed and the watermarked image is lossless. Each feature value is generated by its corresponding gray level of the image; subsequently the feature value is XORed with the embedding bits. In the verification process, the watermark bits can be extracted by majority voting without either the original image or the watermark logo. The experimental results show that the proposed method is robust resisting common nongeometric attacks, and is also strong enough to resist the geometric attacks, while the retrieval watermark logo is still clearly recognizable. The watermarking scheme based on spatial domain is faster than that based on frequency domain since the step of transferring to frequency domain is not needed. In addition to copyright protection, the proposed method can also be applied to image authentication

References

[1] E. Ganic and A. M. Eskicioglu, "Robust

- DWT-SVD domain image watermarking: embedding data in all frequencies " in *Proc. Int. Multimedia Conf. Multimedia and security*, Magdeburg, pp. 166-174, 2004
- [2] I. J. Cox, J. Kilian, T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for images, audio and video," in *Proc. IEEE ICIP*, pp. 243-246, 1996.
- [3] I. J. Cox, J. Kilian, F. T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for multimedia," *IEEE Trans. Image Processing*, vol. 6, pp. 1673-1687, Dec. 1997.
- [4] C. I. Podilchuk and W. Zeng, "Image-adaptive watermarking using visual models," *IEEE J. Sel. Areas Commun.*, vol. 16, pp. 525-539, May 1998.
- [5] D. Y. Chen, O. Ming, and J. L. Wu, "A shift-resisting public watermark system for protecting image processing software," *IEEE Trans. Consumer Electron.*, vol. 46, pp. 404-414, Aug. 2000.
- [6] T. Tachibana, M. Fujiyoshi, and H. Kiya, "An image-quality guaranteed watermarking scheme with spreading spectrum of watermark," in *Proc. IEEE ISCIT*, pp. 330-334, 2004.
- [7] C. F. Wu and W. S. Hsieh, "Image refining technique using digital watermarking," *IEEE Trans. Consumer Electronics*, vol. 46, pp. 1-5, Feb. 2000.
- [8] H. Zhou, C. Qi, and X. Gao, "Low luminance smooth blocks based watermarking scheme in DCT domain," in *Proc. Int. Conf. Communications, Circuits and Systems*, Guilin, pp. 19-23, 2006.
- [9] F. Y. Duan, I. King, L. W. Chan, and L. Xu, "Intra-block algorithm for digital watermarking," in *Proc. Fourteenth Int. Conf. Pattern Recognition*, Brisbane, pp. 1589-1591, 1998.
- [10] O. H. Kwon, Y. S. Kim, and R. H. Park, "A variable block-size dot-based watermarking method," *IEEE Trans. Consum. Electron.*, vol. 45, pp. 1221-1229, Nov. 1999.
- [11] S. Thiemert, T. Vogel, J. Dittmann, and M. Steinebach, "A high-capacity block based video watermark," in *Proc. IEEE EUROMICRO*, pp. 457-460, 2004.
- [12] S. Yang, Z. Lu, and F. Zou, "A robust adaptive image watermarking algorithm," in *Proc. IEEE ICME*, pp. 1723-1726, 2004.
- [13] G. C. Langelaar and R. L. Lagendijk, "Optimal differential energy watermarking of DCT encoded images and video," *IEEE Trans. Image Processing*, vol. 10, pp. 148-158, Jan. 2001.
- [14] S. Kimpan, A. Lasakul, and S. Chitwong,

- "Variable block size based adaptive watermarking in spatial domain," in *Proc. IEEE ISIT*, pp. 374-377, 2004.
- [15] A. K. Mahmood and A. Selin, "Spatially adaptive wavelet thresholding for image watermarking," in *Proc. IEEE ICME*, Toronto, pp. 1597-1600, 2006.
- [16] J. Huang and C. Yang, "Image digital watermarking algorithm using multiresolution wavelet transform," in *Proc. IEEE Int. Conf. Systems, Man and Cybernetics*, , pp. 2977-2982, 2004.
- [17] F. Khelifi, A. Bouridane, F. Kurugollu, and A. I. Thompson, "An improved wavelet-based image watermarking technique," in *Proc. IEEE AVSS*, pp. 588-592, 2005.
- [18] G. Zhang, S. Wang, and Q. Wen, "An adaptive block-based blind watermarking algorithm," in *Proc. IEEE ICSP*, pp. 2294-2297, 2004.
- [19] B. K. Lien and W. H. Lin, "A watermarking method based on maximum distance wavelet tree quantization," in *Proc. 19th Conf. Computer Vision, Graphics and Image Processing*, pp. 269-276, 2006.
- [20] T. H. Chen, G. Horng, and W. B. Lee, "A publicly verifiable copyright-proving scheme resistant to malicious attacks," *IEEE Trans. Industrial Electron.*, vol. 52, pp. 327-334, Feb. 2005.
- [21] D. C. Lou, H. K. Tso, and J. L. Liu, "Robust copyright protection scheme for digital images using the low-band characteristic," *Optical Engineering*, vol. 45, pp. 107002-107006, 2006.
- [22] W. T. Tan and A. Zakhor, "Video multicast using layered FEC and scalable compression," *IEEE Trans. Circuits and Systems for Video Technology*, vol. 11, pp. 373-386, Mar. 2001.
- [23] R. Puri, K. Ramchandran, K. W. Lee, and V. Bharghavan, "Forward error correction (FEC) codes based multiple description coding for internet video streaming and multicast," *Signal Processing: Image Communication*, vol. 16, pp. 745-762, 2001.
- [24] T. Noguchi, M. Yamamoto, and H. Ikeda, "Reliable multicast protocol applied local FEC," in *Proc. IEEE International Conference on Communication*, pp. 2348-2353, 2001.
- [25] J. Rosenberg and A. H. Schulzrinne, "An RTP payload format for generic forward error correction," RFC 2733, 1999.
- [26] Y. Nakayama and S. Aikawa, "Cell discard and TDMA synchronization using FEC in wireless ATM systems," *IEEE Journal on Selected Areas in Communications*, vol. 15, pp. 29-34, Jan. 1997.
- [27] G. M. Davis, J. N. Danskin, and S. Xiyong, "Joint source and channel coding for Internet image transmission," in *Proc. Image Processing*, pp. 21-24, 1996.
- [28] E. Ayanoglu, P. Pancha, A. R. Reibman, and S. Talwar, "Forward error control for MPEG-2 video transport in a wireless ATM LAN," *ACM/Baltzer Mobile Networks and Application*, vol. 1, pp. 245-257, 1996.
- [29] S. Aikawa, Y. Motoyama, and M. Umehira, "Forward error correction schemes for wireless ATM systems," in *Proc. IEEE Personal communications*, pp. 454-458, 1996.
- [30] M. A. Kousa and L. F. Turner, "Multichannel adaptive forward error-correction system," *Communications, Speech and Vision, IEE Proceedings I*, vol. 140, pp. 357-364, Oct. 1993.
- [31] A. J. Viterbi, "Error control for data communication," *ACM SIGCOMM Computer Communication Review* vol. 6, pp. 27-37, Jan. 1976.
- [32] L. Rizzo and L. Vicisano, "RMDP: an FEC-based reliable multicast protocol for wireless environments," in *Proc. ACM SIGMOBILE Mobile Computing and Communications Review*, pp. 23-31, 1998.
- [33] J. Haartsen, M. Naghshineh, and J. Inouye, "Bluetooth: vision, goals, and architecture," *Mobile Computing and Comm. Rev.*, vol. 2, pp. 38-45, 1998.
- [34] The Bluetooth Special Interest Group, "Bluetooth Specification", <http://www.bluetooth.com>.
- [35] J. Haartsen, "Bluetooth-The universal radio interface for ad hoc wireless connectivity," *Ericsson Review*, no 3, pp. 110-117, 1998.
- [36] W. Stallings, *Cryptography and network security*, 4 ed: Prentice Hall, 2005.
- [37] F. A. P. Petitcolas, *Weakness of existing watermark scheme* 1997 [online] Available:<http://www.petitcolas.net/fabien/watermarking/stirmark/index.html>.
- [38] PhotoImpact 11 software, <http://www.ulead.com>, Ulead system, Inc.
- [39] R. C. Gonzalez and R. E. Woods, *Digital image processing* 2ed: Prentice Hall, 2002.